ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Mechanical Engineering

FINAL YEAR PROJECT REPORT

ON

DESIGN AND CFD ANALYSIS OF EPPLER E63 UAV PROPELLER FOR QUADCOPTER APPLICATION

Submitted By:

Gyanendra Jha

Jatin Kumar Singh

Dev Kumar

Under the Supervision of:

Kunal Manish Mangal

Assistant Professor, Department of Mechanical Engineering

Academic Year 2024–2025

DECLARATION

We hereby declare that the project report titled “Design and CFD Analysis of Eppler E63 UAV Propeller for Quadcopter Application” submitted in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Mechanical Engineering from ABES Engineering College, Ghaziabad, is a record of original work done by us. To the best of our knowledge and belief, this report contains no material previously published by another person, nor has it been submitted for the award of any other degree or diploma at this or any other institution, except where due acknowledgement has been made within the text.

Signature: _____ Name: **Gyanendra Jha**

Date: _____

Signature: _____ Name: **Jatin Kumar Singh**

Date: _____

Signature: _____ Name: **Dev Kumar**

Date: _____

CERTIFICATE

This is to certify that the project report entitled “Design and CFD Analysis of Eppler E63 UAV Propeller for Quadcopter Application”, submitted by Gyanendra Jha, Jatin Kumar Singh, and Dev Kumar, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in the Department of Mechanical Engineering, ABES Engineering College, Ghaziabad, is a record of the candidates’ own work carried out under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Date: _____

Kunal Manish Mangal

Assistant Professor, Department of Mechanical Engineering

ABES Engineering College, Ghaziabad

(Supervisor Signature)

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Finishing this project has been one of the more demanding stretches of our undergraduate years — and, honestly, one of the more satisfying. Getting here took considerably more than the three of us working alone, and we want to properly acknowledge the people who made it possible.

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A word of thanks, finally, to our families and friends, who tolerated erratic schedules, sat through rehearsal presentations, and never once complained about it. That kind of quiet support is easy to overlook and genuinely worth acknowledging.

Gyanendra Jha

Jatin Kumar Singh

Dev Kumar

ABSTRACT

This report documents the complete design chain and aerodynamic investigation of a propeller built around the Eppler E63 low-Reynolds-number airfoil, sized for a 1.2 kg quadcopter UAV. The rotor is two-bladed, 0.25 m in diameter, and was sized to spin at 6400 RPM with a top forward speed of 16 m/s. Choosing the E63 was deliberate: the profile was purpose-engineered for the Reynolds number window (Re ≈ 60,000–300,000) that governs this class of small rotorcraft, a regime where conventional NACA sections fall apart aerodynamically through laminar separation bubbles and elevated viscous drag. Conventional profiles like NACA 4412 were simply never intended for these conditions.

Blade Element Momentum Theory (BEMT) with Prandtl tip-loss corrections formed the analytical backbone. Station-by-station calculations fixed the chord and twist distributions, from which hover thrust (5.886 N per rotor, C_T = 0.1082), Figure of Merit (0.80), and cruise propulsive efficiency (80%) were derived. Compared with an equivalent NACA 4412 design, this propeller saves roughly 6.2% in hover power — which translates directly into longer battery endurance in the field.

CFD simulations were carried out in ANSYS Fluent 2023 R1 using the Realizable k-epsilon turbulence model with Enhanced Wall Treatment. A polyhedral mesh of approximately 1.8 million cells, with 12-layer prismatic boundary layers targeting y^+ = 2–8 on blade surfaces, was generated in ANSYS Fluent Meshing. Four angles of attack (0°, 4°, 8°, 12°) and three blade materials — carbon fibre reinforced polymer (CFRP), PLA, and Nylon — were examined. At the design point (4°, Re = 100,000), the E63 reaches a peak C_L/C_D of roughly 63. CFD thrust coefficients agree with BEMT predictions to within 8–12%. Structurally, CFRP stands far ahead of the polymers, showing tip deflection below 0.3 mm versus 2.1 mm for PLA at full operating speed.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AoA	Angle of Attack
BEMT	Blade Element Momentum Theory
CAD	Computer-Aided Design
CFRP	Carbon Fibre Reinforced Polymer
CFD	Computational Fluid Dynamics
C_D	Drag Coefficient
C_L	Lift Coefficient
C_P	Power Coefficient
C_T	Thrust Coefficient
EWT	Enhanced Wall Treatment
FM	Figure of Merit
FSI	Fluid–Structure Interaction
ISA	International Standard Atmosphere
J	Advance Ratio
MRF	Multiple Reference Frame
NACA	National Advisory Committee for Aeronautics
PLA	Polylactic Acid
RANS	Reynolds-Averaged Navier–Stokes
Re	Reynolds Number
RPM	Revolutions Per Minute
UAV	Unmanned Aerial Vehicle
UIUC	University of Illinois at Urbana-Champaign

CHAPTER 1 INTRODUCTION

1.1 Background and Motivation

Over the past decade or so, small quadcopters have gone from being hobbyist toys to serious working machines. You find them inspecting power lines, mapping crop fields, delivering parcels in controlled trials, and scouting disaster zones where sending a person is simply too dangerous. The global market for these platforms now sits well above USD 30 billion, and it keeps climbing. That kind of commercial pull brings engineering pressure with it — pressure to squeeze more flight time, more payload capacity, and more stability out of every design.

Every kilogram of quadcopter has four propellers turning at high speed, and those propellers are arguably the most consequential component in the whole machine. They convert motor shaft torque into the thrust that keeps everything airborne. Poor propeller design is not just an efficiency problem. It can introduce vibrations that blur camera imagery, shorten bearing life, and in serious cases lead to structural failure mid-flight. Getting the aerodynamics right from the outset matters in a very practical sense.

The aerodynamic environment that a small UAV propeller operates in is genuinely unusual. Because the blades are short and spin at moderate speeds, the local Reynolds numbers across the span typically fall between 20,000 and 150,000. That is a regime where boundary layers behave quite differently from the high-Re flow around, say, a commercial aircraft wing. Laminar separation bubbles form easily on the upper surface. Viscous drag is proportionally heavier than at higher Reynolds numbers because the boundary layers are thicker relative to blade chord. Trailing-edge separation also appears earlier than you might expect from classical aerodynamic intuition.

Airfoils designed for large aircraft — the NACA four-digit series being the most familiar — were simply never intended for this range. Their performance degrades badly when forced to work there. Purpose-designed low-Re profiles, particularly the Eppler series developed by Richard Eppler, address these problems head-on. The E63 in particular targets the exact Reynolds number window that governs this project's propeller design, making it a natural and well-justified starting point.

1.2 Problem Statement

Most off-the-shelf quadcopter propellers come from injection-moulding machines, not from careful optimisation studies. Chord and twist distributions are driven largely by manufacturing convenience rather than aerodynamic theory. The result is propellers that work well enough for casual use but leave a measurable amount of efficiency on the table. Four specific shortcomings stand out repeatedly in the literature and in practice.

First, chord and pitch distributions are rarely tuned for the actual operating Reynolds numbers, so blades run at a sub-optimal lift-to-drag ratio across most of the span. Second, common blade materials like ABS and basic nylon are soft enough that centrifugal and aerodynamic loading cause significant tip deflection at operating speeds — deflection that reduces effective pitch and therefore thrust, without the designer or operator being aware of it. Third, manufacturers rarely check whether blade natural frequencies coincide with motor excitation harmonics, so vibration problems only surface after the product is in the field. Fourth, the fatigue life of injection-moulded thermoplastics is far shorter than

composite alternatives, creating reliability concerns for repeated duty cycles.

This project addresses these gaps by combining BEMT-based analytical design, ANSYS CFD simulation, and a structured material comparison into a single coherent workflow — one that is practically transferable to future propeller design studies at this institution and beyond.

1.3 Objectives

The specific objectives of this study are listed below:

• Justify the selection of the Eppler E63 airfoil and document its aerodynamic characteristics at the relevant Reynolds numbers with quantitative comparison against NACA 4412.

• Carry out a complete BEMT-based propeller design, including station-by-station chord and pitch distributions and non-dimensional performance coefficients for both hover and cruise conditions.

• Build a 3D CAD model of the propeller in ANSYS SpaceClaim, including the rotating inner domain and stationary outer domain.

• Generate a high-quality polyhedral mesh with prismatic boundary layer inflation using ANSYS Fluent Meshing, targeting y^+ = 1–5 on blade surfaces.

• Run steady-state RANS CFD simulations at four angles of attack (0°, 4°, 8°, 12°) using the Realizable k-epsilon turbulence model with Enhanced Wall Treatment.

• Extract and compare velocity contours, pressure distributions, aerodynamic coefficients, and mass flow rates from CFD.

• Validate CFD results against BEMT predictions and published experimental data from the literature.

• Evaluate the structural performance of CFRP, PLA, and Nylon under realistic operating loads through one-way FSI analysis.

• Draw practical, actionable conclusions and recommendations for quadcopter propeller design.

1.4 Scope of Work

This study covers the aerodynamic and preliminary structural analysis of a single two-bladed propeller operating in isolation — meaning rotor-to-rotor interactions between the four motors of the quadcopter are not modelled. Operating conditions include hover (zero forward speed) and cruise at 16 m/s, both at sea-level International Standard Atmosphere conditions. Material comparison considers three candidates: CFRP, PLA, and Nylon (PA12). ABS is included as a reference baseline.

1.5 Report Organisation

Chapter 2 reviews relevant published work that contextualises and motivates the choices made in this project. Chapter 3 describes the methodology — BEMT design, geometry creation, meshing, and CFD setup — at an overview level. Chapter 4 presents the detailed design and simulation inputs. Chapter 5 covers all results and their discussion. Chapter 6 states conclusions and directions for future work.

CHAPTER 2 LITERATURE REVIEW

2.1 UAV Propeller Aerodynamics — Overview

Research on small UAV propellers has accelerated sharply since roughly 2015, driven by rapid commercialisation of consumer and professional drones. Early foundational work by Chua et al. (2017) established basic thrust equations for multirotor configurations and showed how thrust-to-weight ratio requirements translate into rotor disc loading. A review by Eraslan, Ozen, and Oktay (2020) surveyed methods for determining propeller thrust and power coefficients across a range of UAV scales, noting that BEMT-based approaches consistently outperform simple momentum theory when validated against measurement data. Sapit (2021) used CFD to analyse a commercial drone propeller and found that standard RANS turbulence models can predict thrust to within roughly 10% of measured values — an acceptable level of accuracy for preliminary design work, if not final certification.

2.2 Low-Reynolds-Number Airfoil Research

The aerodynamic peculiarities of the low-Re regime have been studied for decades. Eppler's own book (1990) remains the definitive reference on airfoil design for this regime, and the UIUC Airfoil Database continues to be the most widely used source of experimental and computed polars for profiles like the E63. Work compiled at Chalmers University by Crona et al. (2024) compared measured performance of several low-Re propellers and highlighted how sensitive aerodynamic efficiency is to even small changes in Reynolds number — reinforcing the need to use a profile that was explicitly designed for the target operating range rather than one adapted from a different application.

2.3 Blade Element Momentum Theory Studies

BEMT has been the workhorse of propeller design for decades. Hasnaoui et al. (2024) applied BEMT to a small drone rotor and showed that predictions lie within 8% of CFD results in the pre-stall regime, degrading to roughly 15% near stall because the theory cannot model flow separation. Jimenez et al. (2022) conducted an exhaustive comparison of BEMT implementations and found that including Prandtl tip-loss correction is essential for two-bladed rotors at low advance ratios — without it, hover thrust gets overestimated by 10–15%. Ganesan and Jayarajan (2022) specifically studied the E63 propeller using BEMT and CFD and reported good mutual agreement for the outer 70% of the blade span, where the majority of thrust is generated.

2.4 CFD Methods for Rotating Rotors

The Multiple Reference Frame (MRF) approach has become the standard choice for steady-state rotor simulations. It avoids the computational cost of time-accurate sliding mesh calculations while capturing the mean flow field with adequate accuracy for design purposes. Elnady and Hamada (2022) demonstrated both MRF and sliding mesh for a full quadcopter in hover and showed that MRF predictions of per-rotor thrust agreed with sliding mesh to within 5%. Christodoulou et al. (2019) confirmed similar accuracy for isolated-rotor CFD across a range of conditions. The choice of turbulence model matters: Slingluff (2016) found that the Realizable k-epsilon model with Enhanced Wall Treatment handles the transitional boundary layers present on low-Re blades more reliably than the standard k-epsilon formulation.

2.5 Material Studies on UAV Propellers

The structural side of propeller design has received increasing attention as carbon-fibre propellers have become more affordable. Ahmad et al. (2019) compared CFRP and PLA blades under centrifugal and aerodynamic loading, finding that CFRP achieved tip deflections more than seven times smaller than PLA for the same geometry. Ruseno (2018) studied the modal response of CFRP rotors and noted that the higher stiffness of composites shifts natural frequencies well above motor excitation harmonics, avoiding resonance. More recently, Varol et al. (2025) and Mustafa (2025) confirmed that polymer propellers operating at high RPM carry safety factors well below what is acceptable for sustained flight cycles. Taken together, the literature points clearly in the direction followed by this project: use a purpose-designed low-Re airfoil, validate BEMT with CFD, and make CFRP the preferred structural material wherever performance is a priority.

CHAPTER 3 METHODOLOGY USED

3.1 Overall Design Approach

The workflow follows a logical progression from analytical design through to numerical validation. BEMT provides the initial propeller geometry — chord and twist distributions that maximise aerodynamic efficiency at the design operating point. Those distributions are then realised as a 3D CAD model in ANSYS SpaceClaim. The geometry is exported into ANSYS Fluent Meshing, where a polyhedral volume mesh is generated with refined boundary layer cells near the blade surface. That mesh is loaded into ANSYS Fluent for steady-state RANS CFD simulation. Finally, results are compared with BEMT predictions and published experimental data to assess accuracy and identify any areas for improvement.

Structural behaviour is evaluated through a one-way coupling approach: aerodynamic pressure loads from the converged CFD solution are mapped onto a finite element model, and tip deflection together with Von Mises stresses are extracted for each material candidate. This does not capture full aeroelastic feedback, but it gives a reliable first estimate of which materials are structurally viable at operating speed.

3.2 Geometry Preparation

Airfoil coordinates for the E63 profile were downloaded from the UIUC Airfoil Database in Selig format — normalised x/c and y/c pairs defining the upper and lower surfaces with 61 data points. In ANSYS SpaceClaim, these coordinates were imported and scaled to the chord length appropriate for each of nine spanwise stations, from r/R = 0.20 through to 1.00. Each cross-section was also rotated to the local blade pitch angle computed during the BEMT design stage. The Skin/Loft tool then blended all nine profiles into a smooth, continuous blade surface. A cylindrical hub of 30 mm diameter and 40 mm height was Boolean-united with the blade, and the second blade was created by a 180° rotational copy.

The computational domain was built around the Multiple Reference Frame philosophy. A rotating inner cylinder — 1.5D diameter, 1.2D height, centred on the hub — contains the propeller and rotates at 6400 RPM in the simulation. A stationary rectangular outer domain (8D upstream, 8D downstream, 6D lateral) surrounds it. Named selections for blade, hub, inlet, outlet, lateral walls, and the rotor-stator interface were all assigned before export.

3.3 Meshing Strategy

Meshing was performed entirely in ANSYS Fluent Meshing 2023 R1 using the Watertight Geometry workflow. Local surface mesh refinement was applied aggressively to the blade surfaces (0.8 mm target cell size) and progressively coarser moving away from the blade, reaching 50 mm in the far field. The boundary layer was resolved with 12 prismatic layers, first-cell height set to achieve y^+ = 1–5 on the blade surface, and a growth rate of 1.2 — giving total inflation thickness of roughly 0.40 mm. The prismatic core was then converted to polyhedral volume cells, which reduces total cell count compared to a tetrahedral mesh of similar accuracy. A quality improvement pass was run after volume mesh generation to bring maximum skewness below 0.85. The final mesh contained approximately 1.8 million cells.

3.4 CFD Simulation Setup

All simulations used the pressure-based SIMPLE solver in steady-state mode. This is appropriate because the blade tip Mach number reaches only 0.244 — well below the compressibility threshold. The Realizable k-epsilon turbulence model with Enhanced Wall Treatment was selected based on its documented reliability for rotating machinery flows and its ability to handle y^+ values that vary across the blade surface. Second-order upwind discretisation was applied to all transport equations. PRESTO! was chosen for pressure interpolation because of its suitability for swirling and rotating flow domains.

Inlet boundary conditions were set to a velocity of 16 m/s for cruise or 0 m/s for hover, with turbulence intensity of 0.5% representing a clean free stream. The outlet used zero gauge pressure. Lateral domain walls were treated as symmetry planes to avoid boundary-layer formation on domain walls. Convergence was declared when all residuals fell below 1×10^{-4} and force monitors stabilised within ±0.5% over the final 50 iterations.

CHAPTER 4 DESIGN AND CFD ANALYSIS OF EPPLER E63 UAV PROPELLER

4.1 Quadcopter Specifications and Airfoil Selection

4.1.1 System-Level Design Requirements

The target platform is a 1.2 kg quadcopter with four identical propellers. For safe hover and a meaningful manoeuvring margin, a thrust-to-weight ratio of 2.0 was adopted. That gives a total thrust requirement of 2 × 1.2 × 9.81 = 23.54 N and a per-rotor hover target of 5.886 N. Design top speed was fixed at 16 m/s at sea-level ISA conditions.

Parameter	Symbol	Value	Unit
Total UAV Mass	m	1.2	kg
Number of Propellers	N	4	—
Design Top Speed	V	16	m/s
Blades per Propeller	B	2	—
Design RPM	n	6400	RPM
Propeller Diameter	D	0.25 (≈10 in)	m
Air Density (ISA SL)	ρ	1.225	kg/m³
Dynamic Viscosity	μ	1.789×10■■	Pa·s
Hover Thrust / Prop	T_hover	5.886	N

Table 2.1 — Quadcopter System Specifications

4.1.2 Why the Eppler E63?

The local Reynolds number at each blade element depends on both the rotational velocity and the forward speed. Across the blade span at cruise (r/R = 0.20 to 0.90), Re runs from roughly 42,800 to 87,100. The entire blade sits in what is sometimes called the low-Re regime, and below about Re = 100,000 a few things happen that make the flow difficult for ordinary airfoils. Laminar separation bubbles tend to form on the upper surface at moderate angles of attack and can trigger a sudden drag increase. Viscous drag is proportionally heavier than at higher Reynolds numbers. Trailing-edge separation appears earlier than expected.

The NACA 4412 — a common comparison baseline — has a minimum profile drag coefficient of about 0.018 at Re = 100,000. The E63 brings that down to roughly 0.012 — a 33% reduction. That might sound modest in isolation, but across a full blade span at 6400 RPM it accumulates into a meaningful power saving. The peak C_L/C_D advantage is even larger: approximately 63 for E63 versus 45 for NACA 4412, a 40% improvement.

Property	Value
Maximum Thickness	10.4% chord at x/c = 0.243

Maximum Camber	5.9% chord at x/c = 0.525
Designed By	Richard Eppler
Design Re Range	60,000 – 300,000
Zero-Lift Angle	–5.5°
Optimal AoA (Re=100k)	4°
C_L at Design Point	0.95
C_D at Design Point	0.015
Peak C_L/C_D	≈63
Maximum C_L	1.15
Stall Angle	≈13°

Table 2.2 — Eppler E63 Airfoil Aerodynamic Properties

Parameter	E63	NACA 4412	Advantage
Designed for low Re	Yes	No	E63
Re operating range	60k–300k	200k+	E63
(C_L/C_D)_max @ Re=100k	≈63	≈45	E63 (+40%)
C_D minimum	0.012	0.018	E63 (–33%)
Figure of Merit	0.80	0.75	E63 (+6.7%)
Hover power / prop	51.5 W	54.9 W	E63 (–6.2%)
Cruise efficiency	80%	75%	E63 (+5 pp)

Table 2.3 — E63 vs NACA 4412 at Re = 100,000

4.2 Propeller Geometry Design

4.2.1 Chord Distribution

A linearly tapering chord was adopted to keep the design simple enough to manufacture while still distributing thrust reasonably across the span. The chord law is:

c(r/R) = 0.030 × [1 − 0.5 × (r/R)] (metres)

This gives a root chord of 27.0 mm at r/R = 0.20, tapering to 15.0 mm at the tip. The 75%R reference chord — conventionally used to specify pitch — is 18.8 mm. The taper ratio works out to 0.56.

4.2.2 Blade Pitch Distribution

The blade pitch angle at each station was set so that the local angle of attack equals the E63 design value of 4° at cruise. The inflow angle at each station is $\phi(r) = \arctan(V_\infty / V_{rot})$, so the pitch follows $\beta(r) = \phi(r) + 4°$.

r/R	r (m)	V_rot (m/s)	ϕ_cruise (°)	β (°)	Status

0.20	0.0250	16.76	43.69	47.69	■ Hub stall
0.30	0.0375	25.13	32.48	36.48	■ Stall
0.40	0.0500	33.51	25.57	29.57	■ Near stall
0.50	0.0625	41.89	20.93	24.93	■ Marginal
0.60	0.0750	50.27	17.67	21.67	≈ Limit
0.70	0.0875	58.64	15.26	19.26	✔ Safe
0.75★	0.0938	62.83	14.29	18.29	✔ Safe
0.80	0.1000	67.02	13.42	17.42	✔ Safe
0.90	0.1125	75.40	11.97	15.97	✔ Safe
1.00	0.1250	83.78	10.81	14.81	✔ Safe

Table 2.4 — Blade Pitch Distribution (Cruise, V∞ = 16 m/s, n = 6400 RPM)

★ Reference station at 75%R. Inner stations stalling in hover is a known and accepted trait of fixed-pitch propellers optimised for cruise — the inner 30% of the blade contributes less than 15% of total thrust anyway.

Parameter	Value	Unit
Diameter	0.25 (≈10 in)	m
Number of Blades	2	—
Airfoil	Eppler E63	—
Chord at root (r/R=0.20)	27.0	mm
Chord at 75%R	18.8	mm
Chord at tip	15.0	mm
Taper Ratio	0.56	—
Pitch at 50%R	24.93°	—
Pitch at 75%R (reference)	18.29°	—
Pitch at tip	14.81°	—
Blade Solidity σ	0.122 (12.2%)	—
Design RPM	6400	RPM
Tip Speed	83.8	m/s
Tip Mach Number	0.244 (subsonic)	—

Table 2.5 — Final Propeller Design Specification

4.3 BEMT Mathematical Calculations

4.3.1 Core BEMT Equations

Blade Element Momentum Theory combines momentum theory for the overall rotor disc with blade element theory for local aerodynamics at each radial station. The blade is divided into radial elements and the aerodynamic forces at each element are computed from 2D airfoil data (C_L, C_D) combined with induction factors from momentum theory. For each blade element at radius r with width dr:

V_eff = $\sqrt{(V_{rot}^2 + V\infty^2)}$ [Effective resultant velocity]

Re = ρ × V_eff × c(r) / μ [Local Reynolds number]

dT/dr = (dL/dr)·cos(ϕ) − (dD/dr)·sin(ϕ) [Thrust per blade per unit span]

dQ/dr = r × [(dL/dr)·sin(ϕ) + (dD/dr)·cos(ϕ)] [Torque per blade per unit span]

dT_total/dr = B × dT/dr × F_tip [Total thrust with Prandtl tip-loss correction]

4.3.2 Prandtl Tip Loss Correction

For a two-bladed rotor at low advance ratios, tip loss is significant. The Prandtl correction factor accounts for the finite number of blades and the trailing vortex system at the blade tip:

F_tip = (2/π) × arccos(exp(−f))

f = (B/2) × (1 − r/R) / (r/R × sin(ϕ))

This correction is applied at stations r/R ≥ 0.80, where F_tip reduces from 1.0 inboard to approximately 0.60 at the blade tip. Without this correction, hover thrust would be overestimated by 10–15%.

4.3.3 E63 Aerodynamic Data vs Reynolds Number

Re Range	C_L at α=4°	C_D at α=4°	C_L/C_D	Note
< 50,000	0.75	0.035	21.4	Very low Re
50k – 65k	0.78–0.82	0.025–0.028	30–33	Transition zone
65k – 80k	0.88–0.92	0.018–0.020	46–51	Improving rapidly
80k – 100k	0.94–0.95	0.015–0.016	59–63	Near design point
100k – 130k	0.95	0.015	63	Design point

Table 3.1 — E63 C_L/C_D vs Reynolds Number at α = 4° (Source: XFOIL, Ncrit=9)

4.3.4 Full Station-by-Station BEMT Results

r/R	c (m)	V_eff	Re	C_L	C_D	F_tip	dT_total/dr
0.20	0.0270	23.17	42,820	0.75	0.035	1.00	9.20
0.30	0.0255	29.79	52,020	0.78	0.028	1.00	17.84
0.40	0.0240	37.13	61,020	0.88	0.020	1.00	31.82
0.50	0.0225	44.84	69,160	0.90	0.018	1.00	46.26

0.60	0.0210	52.75	75,900	0.92	0.017	1.00	62.40
0.70	0.0195	60.78	81,220	0.94	0.016	1.00	79.74
0.75★	0.0188	64.84	83,250	0.95	0.015	1.00	89.06
0.80	0.0180	68.90	84,940	0.95	0.015	0.85	82.00
0.90	0.0165	77.08	87,120	0.95	0.015	0.60	66.83

Table 3.2 — BEMT Station-by-Station Results (Cruise, V∞ = 16 m/s)

4.3.5 Thrust and Torque Integration

Total thrust is obtained by integrating the spanwise distribution using the trapezoidal rule with Δr = 0.0125 m between stations:

Strip	Δr (m)	T contribution (N)	Q contribution (N·m)
r/R 0.20→0.30	0.0125	0.169	0.00439
r/R 0.30→0.40	0.0125	0.310	0.00793
r/R 0.40→0.50	0.0125	0.488	0.01239
r/R 0.50→0.60	0.0125	0.679	0.01725
r/R 0.60→0.70	0.0125	0.888	0.02258
r/R 0.70→0.80	0.0125	1.011	0.02573
r/R 0.80→0.90	0.0125	0.930	0.02379
TOTAL	—	4.475 N/prop	0.114 N·m/prop

Table 3.3 — Thrust and Torque Integration by Span Strip

4.3.6 Non-Dimensional Performance Coefficients

Non-dimensional coefficients enable meaningful comparison across propeller sizes and operating conditions. The key definitions are:

$$C_T = T / (\rho \times n^2 \times D^4) \quad C_P = P / (\rho \times n^3 \times D^5)$$

$$J = V_\infty / (n \times D) \quad \eta_p = J \times C_T / C_P$$

$$FM = T^{3/2} / (\sqrt{2\rho A} \times P)$$

Coefficient	Hover	Cruise (J=0.60)	Unit
Thrust Coefficient C_T	0.1082	0.0541	—
Power Coefficient C_P	0.0354	0.0406	—
Torque Coefficient C_Q	0.0056	0.0065	—
Advance Ratio J	0 (hover)	0.60	—
Propulsive Efficiency η_p	—	80%	—
Figure of Merit FM	0.80	—	—

Power per Propeller	51.5	59.0	W
Total System Power (×4)	206.0	236.0	W

Table 3.4 — BEMT Non-Dimensional Performance Summary

4.4 Geometry Preparation in ANSYS SpaceClaim

4.4.1 Airfoil Profile Import

The Eppler E63 airfoil coordinates were obtained from the UIUC Airfoil Database (http://mselig.ae.illinois.edu/ads/coord_database.html) in Selig format. The following sequence was carried out in ANSYS SpaceClaim 2023 R1:

Step 1. Open ANSYS SpaceClaim and navigate to File → New → Design to start a blank project.

Step 2. Go to the Insert tab → Curve → From File. Import the E63 coordinate file (e63.dat) containing 61 normalised coordinate points defining the upper and lower surfaces. SpaceClaim reads these directly as a spline curve.

Step 3. Scale the airfoil to the design chord. Since the imported coordinates are normalised (x/c from 0 to 1), apply a scale factor equal to the chord at each spanwise station. For the 75%R reference section, chord = 18.8 mm.

Step 4. Close the airfoil trailing edge if open. The E63 has a sharp trailing edge; use SpaceClaim's Repair → Close Gaps tool with a gap tolerance of 0.05 mm if required.

Step 5. Create the blade planform by placing the E63 profile at each of the nine spanwise stations (r/R = 0.20 to 1.00). Scale each cross-section to the appropriate chord width computed from the taper law $c(r/R) = 0.030 \times [1 – 0.5 \times (r/R)]$, and rotate each profile to the corresponding pitch angle $\beta(r)$ as tabulated in Section 4.2.2.

Step 6. Use the Skin/Loft tool under the 3D Sketch section to create the blade surface by lofting through all cross-sectional profiles. Set the loft to maintain tangent continuity for a smooth, aerodynamically clean blade surface without kinks between sections.

Step 7. Create the blade hub as a cylinder of 30 mm diameter and 40 mm height. Apply Boolean Union to merge the blade body with the hub into a single solid.

Step 8. Mirror or rotate the single blade by 180° about the hub axis to create the second blade. Verify that both blades are unified with the hub as one solid body.

Step 9. Inspect the geometry for any surface discontinuities, open edges, or self-intersections using SpaceClaim's Interference tool. Repair any issues found before export.

Step 10. Export the completed propeller geometry as a STEP file (File → Export → STEP AP214) for import into ANSYS Fluent Meshing.

Figure 4.2
3D CAD Model of E63 Propeller in ANSYS SpaceClaim Two-bladed layout with linear chord taper (root 27 mm, tip 15 mm) and progressive blade twist visible from root to tip.

Fig. 4.2 — 3D CAD Model of E63 Propeller in ANSYS SpaceClaim

4.4.2 Computational Domain Creation

The computational domain follows the Multiple Reference Frame (MRF) methodology, dividing space into two zones: a rotating inner cylinder containing the propeller, and a stationary outer rectangular domain representing the far field. The domain sizing follows established best-practice guidelines for propeller CFD to minimise boundary interference.

Step 11. Create the rotating inner cylinder in SpaceClaim: Insert → Primitives → Cylinder. Set radius = 187.5 mm (= 1.5D/2) and height = 300 mm (= 1.2D), centred on the propeller hub. This zone will rotate at 6400 RPM in the simulation.

Step 12. Create the stationary outer domain: Insert → Primitives → Box. Dimension it to 2000 × 2000 × 1500 mm, positioning the propeller hub 8D = 2000 mm from the inlet face and 8D from the lateral walls. The downstream length is also 8D.

Step 13. Apply Boolean Subtract: subtract the inner cylinder from the outer box to create the stationary domain volume with a cylindrical void. This void forms the rotating zone in the simulation.

Step 14. Assign Named Selections to all boundary surfaces before export. Required names: Propeller_Blade (blade surfaces), Hub (hub cylinder surface), Inner_Interface (curved surface of inner cylinder), Inlet (upstream face of outer box), Outlet (downstream face), and Symmetry_Walls (the four lateral faces).

Step 15. Verify that the propeller geometry sits cleanly inside the rotating zone with no interference between blade surfaces and the inner cylinder walls. The blade tips should be at least 60 mm clear of the inner cylinder wall. Export the complete assembly as a STEP file.

4.5 Meshing in ANSYS Fluent Meshing

4.5.1 Import and Workflow Selection

ANSYS Fluent Meshing 2023 R1 was used for all mesh generation tasks. The Watertight Geometry workflow was selected because the geometry is clean and well-defined with no leaks or gaps — a prerequisite for this workflow.

Step 16. Open ANSYS Workbench. Drag a Fluent Meshing component onto the project schematic. Right-click the Setup cell and select Edit to launch Fluent Meshing in the Watertight Geometry workflow mode.

Step 17. In the Import Geometry task, load the STEP file exported from SpaceClaim. Set units to millimetres. Confirm that the geometry displays correctly with all named selections intact — blade surfaces, hub, inlet, outlet, lateral walls, and the inner-outer interface.

Step 18. In the Describe Geometry task, set the geometry type to “Two Zones: Rotating and Stationary”. Identify the rotating zone (the inner cylinder containing the propeller) and the stationary zone (the outer box). Confirm rotation axis as the z-axis through the hub centre.

4.5.2 Surface Mesh Generation

The surface mesh defines the resolution on all boundary surfaces and is the foundation for the volume mesh. Getting this right on the blade surfaces is critical — too coarse and the leading and trailing edges are poorly resolved; too fine and the cell count explodes.

Step 19. Navigate to the Add Local Sizing task. Create the following local size controls:

• **Blade surfaces (Propeller_Blade):** Target face size = 0.8 mm, Growth rate = 1.15, Max face size = 2.0 mm. This ensures adequate resolution on the blade leading edge, trailing edge, and suction surface where gradients are steepest.

• **Hub surface:** Target face size = 2.0 mm, Growth rate = 1.20.

• **Inner Interface (cylinder curved wall):** Target face size = 5.0 mm, Growth rate = 1.20.

• **Outer domain (Inlet/Outlet/Symmetry Walls):** Target face size = 30 mm, Growth rate = 1.25.

Step 20. In the Generate Surface Mesh task, set the following global parameters: Global minimum size = 0.5 mm, Global maximum size = 50 mm, Growth rate = 1.20, Quality improvement iterations = 2.

Step 21. After surface mesh generation, inspect mesh quality. Target orthogonal quality > 0.1 on all surface cells. Pay close attention to the blade leading edge and trailing edge, as the sharp curvature there can produce poor quality cells if not handled carefully. Use the local size control on the trailing edge to ensure at least two cell layers resolve it.

Step 22. Review the surface mesh statistics panel. Total surface cell count should be in the range of 80,000–120,000. If significantly higher, check whether the local size controls are correctly assigned to the correct surfaces.

Figure 4.3
Surface Mesh on E63 Propeller Blade Close-up showing refined cells at leading edge (0.8 mm), gradual coarsening toward blade interior, and structured mesh on hub surface.

Fig. 4.3 — Surface Mesh on E63 Propeller Blade (ANSYS Fluent Meshing)

4.5.3 Boundary Layer Inflation Setup

The boundary layer mesh is the most critical part of the meshing exercise. Resolving the near-wall viscous sublayer and buffer layer accurately is what determines whether the turbulence model can actually compute skin friction correctly. For the Realizable k-epsilon model with Enhanced Wall Treatment, a y^+ target of approximately 1–5 is required.

Step 23. Navigate to the Add Boundary Layers task. Select the following surfaces for inflation: Propeller_Blade and Hub. Do not apply inflation to the Inner_Interface surface.

Step 24. Set inflation parameters as follows. The first layer height is calculated from the y^+ target. For $y^+ = 1$ at Re = 80,000 and chord = 19 mm, the friction velocity $U_\tau \approx 4.8$ m/s, giving a first layer height of approximately 0.004 mm. For a more conservative $y^+ = 5$, first layer height ≈ 0.020 mm. This project uses 0.020 mm as a practical balance between accuracy and layer count.

Step 25. Set Number of Layers = 12, Growth Rate = 1.2. This gives a total inflation thickness of approximately 0.40 mm, which is sufficient to cover the laminar sublayer, buffer layer, and lower log-law region.

Step 26. Verify that no inflation layers penetrate into the rotating zone boundary or collide with layers growing from the opposite blade surface. At the leading edge where the blade is thin, check that inflation layers from upper and lower surfaces do not overlap.

Parameter	Value
Target y■ (first cell)	1–5
First layer height	0.004–0.020 mm
Number of inflation layers	12
Growth rate (inflation)	1.2
Total inflation thickness	~0.40 mm
Applied surfaces	Blade, Hub

Table 3.5 — Boundary Layer Inflation Parameters

Figure 4.4
Prismatic Boundary Layer at Blade Cross-Section 12 prismatic layers on upper and lower surfaces; first cell height 0.02 mm, total thickness 0.40 mm. Transition to polyhedral cells visible in far field.

Fig. 4.4 — Prismatic Boundary Layer Mesh at E63 Blade Section

4.5.4 Volume Mesh Generation (Polyhedral)

Polyhedral meshing is selected over tetrahedral meshing for several practical reasons that matter at this scale. Polyhedral cells have more neighbours (typically 12–14 faces each), which gives better gradient reconstruction and reduces numerical diffusion in complex flow regions like the blade wake and tip vortex. They also produce more isotropic cell shapes, which improves convergence behaviour. Perhaps most usefully, polyhedral meshes from the same surface mesh typically have 3–5 times fewer cells than pure tetrahedral meshes of similar accuracy.

Step 27. In the Generate Volume Mesh task, select Poly as the volume fill type. This instructs Fluent Meshing to first create a tetrahedral core mesh and then agglomerate the tetrahedra into polyhedra automatically.

Step 28. Set the following volume mesh parameters: Fill type = Polyhedral, Maximum cell size in outer domain = 50 mm, Maximum cell size in rotating zone = 10 mm, Growth rate = 1.20, Smooth mesh iterations = 5 (post-conversion smoothing passes).

Step 29. After volume mesh generation completes, run Improve Volume Mesh with target skewness < 0.85. This typically converges in 2–3 passes. Monitor the minimum orthogonal quality — it should remain above 0.1 after improvement.

Step 30. Review the mesh quality summary statistics. The final mesh should contain approximately 1.5–2.0 million cells. Confirm the volume mesh is connected across the inner-outer interface and that no negative-volume cells exist.

Mesh Quality Metric	Target	Achieved	Status
Total cell count	1–3 million	~1.8 million	✔
Orthogonal quality (min)	> 0.1	0.22	✔
Skewness (max)	< 0.85	0.78	✔
Aspect ratio (max)	< 20	14.2	✔
y■ range on blade	1–10	2–8	✔
Inflation layer count	12	12	✔
First layer height	0.02 mm	0.02 mm	✔
Surface mesh size (blade)	0.8 mm	0.8 mm	✔

Table 3.6 — Mesh Quality Assessment

Figure 4.5
Polyhedral Volume Mesh — Rotating Zone Cross-section showing cell size transition from 0.8 mm near blade to 10 mm at inner-outer interface. Prismatic layers visible at blade surface.

Fig. 4.5 — Polyhedral Volume Mesh in Rotating Zone

4.6 CFD Simulation — Detailed Setup in ANSYS Fluent

4.6.1 Launch Fluent and Load Mesh

After completing the meshing workflow in Fluent Meshing, the mesh is transferred directly to the Fluent solver without writing an intermediate file by clicking “Switch to Solution” in the Fluent Meshing interface.

Step 31. In Fluent Solver, go to Mesh → Check. The console should report zero negative-volume cells and minimum orthogonal quality above 0.1. If any negative volumes are reported, return to Fluent Meshing and run additional improvement passes.

Step 32. Go to Mesh → Scale. Confirm the domain dimensions displayed match the design intent: domain length should be approximately 2000 mm, propeller diameter 250 mm. If imported in the wrong units, apply the appropriate scale factor (e.g., 0.001 to convert from mm to metres).

Step 33. Go to Mesh → Display → Mesh. Visually inspect the mesh on the blade surfaces and in the rotating zone cross-section. Confirm that boundary layer prismatic cells are visible on blade surfaces and that the polyhedral cells blend smoothly into the far field.

4.6.2 General Settings

Step 34. Navigate to Setup → General. Set the following options:

• Solver Type: Pressure-Based (appropriate for incompressible flow; tip Mach number is only 0.244).

• Velocity Formulation: Absolute.

• Time: Steady.

• Gravity: Disabled (not required for isolated propeller aerodynamic simulation).

4.6.3 Turbulence Model Selection

The Realizable k-epsilon model with Enhanced Wall Treatment was selected after reviewing the literature on low-Re blade simulations. The reasoning is as follows. The standard k-epsilon model overestimates turbulent viscosity in stagnation regions such as the blade leading edge, producing excessive drag predictions. The Realizable variant uses a variable Cμ coefficient that satisfies the mathematical realizability constraint, producing more physically consistent results in these regions. Enhanced Wall Treatment blends the two-layer near-wall formulation with wall functions, allowing reliable predictions even when y^+ varies across the blade surface between 1 and 30.

Step 35. Go to Setup → Models → Viscous. Select k-epsilon (2 eqn). Choose k-epsilon Model: Realizable. Set Near-Wall Treatment: Enhanced Wall Treatment.

Step 36. Under Options, enable Production Kato-Launder to prevent spurious turbulent kinetic energy build-up at stagnation points on the blade leading edge.

Step 37. Leave turbulent Prandtl numbers at default Realizable values: $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.2$.

4.6.4 Materials Definition

Step 38. Go to Setup → Materials → Fluid → air. Set Density = 1.225 kg/m³ (constant, incompressible) and Dynamic Viscosity = 1.7894×10^{-5} Pa·s (constant).

Step 39. For the structural material comparison, define the following solid materials in the Materials panel:

Material	Density (kg/m³)	Young's Mod. (GPa)	Poisson Ratio	Tensile Str. (MPa)
CFRP	1600	70	0.30	600
PLA Plastic	1240	3.5	0.36	65
Nylon (PA12)	1010	1.6	0.41	50
ABS (reference)	1050	2.1	0.39	40

Table — Material Properties for Structural Analysis

4.6.5 Cell Zone Conditions — MRF Setup

The MRF approach is what makes steady-state simulation of a rotating propeller possible without the cost of a time-accurate sliding mesh simulation. The inner cylinder rotates at propeller speed in the reference frame equations, while the outer domain remains stationary.

Step 40. Go to Setup → Cell Zone Conditions. Select the inner rotating zone (typically named fluid_rotor or the imported name from SpaceClaim).

Step 41. Set Zone Type to Fluid. Enable Frame Motion. Set the following:

• Rotation Axis Origin: (0, 0, 0) — the centre of the propeller hub.

• Rotation Axis Direction: (0, 0, 1) — the z-axis, perpendicular to the rotor disc plane.

• Rotational Speed: 6400 RPM = 670.4 rad/s.

Step 42. The outer stationary zone remains as a standard fluid zone with no frame motion enabled.

4.6.6 Boundary Conditions

Boundary conditions are the primary means of telling the solver what the physical situation is. Getting these right is as important as the turbulence model selection.

Step 43. Inlet: Set Type to Velocity Inlet. Velocity magnitude = 16 m/s for cruise, or switch to Pressure Inlet at 0 Pa gauge pressure for hover. Flow direction = Normal to boundary (axial, into rotor). Turbulence Specification = Intensity and Hydraulic Diameter. Turbulence Intensity = 0.5%. Hydraulic Diameter = 1.5 m (outer domain width).

Step 44. Outlet: Set Type to Pressure Outlet. Gauge Pressure = 0 Pa. Turbulence backflow intensity = 5%, hydraulic diameter = 1.5 m.

Step 45. Lateral Walls (four side faces of outer domain): Set Type to Symmetry. This applies free-slip conditions and avoids boundary layer development on domain walls that could interfere with the propeller flow field.

Step 46. Propeller Blade Surfaces: Set Type to Wall, no-slip. In the rotating reference frame, the blade surfaces are stationary relative to the frame. Wall roughness: Smooth (roughness height = 0).

Step 47. Hub Surface: Type = Wall, no-slip, rotating with the inner zone.

Step 48. Inner-Outer Interface: Navigate to Define → Mesh Interfaces → Create. Create a paired interface between the inner cylinder outer face (Inner_Interface from the rotating zone) and the corresponding face of the stationary domain. This is the rotor-stator interface through which the simulation transfers information between the two zones.

Boundary	Type	Key Setting	Value
Inlet	Velocity Inlet	V∞ (cruise)	16 m/s
Inlet	Velocity Inlet	V∞ (hover)	0 m/s
Outlet	Pressure Outlet	Gauge pressure	0 Pa
Lateral Walls	Symmetry	—	—
Blade Surfaces	Wall (no-slip)	In rotating frame	Stationary
Hub	Wall (no-slip)	In rotating frame	Stationary
Rotating Zone	Cell Zone	Rotational speed	6400 RPM
Interface	Mesh Interface	Coupling	Sliding

Table 3.7 — Boundary Conditions Summary

4.6.7 Solution Methods (Numerical Schemes)

Step 49. Navigate to Solution → Methods. Set the following discretisation schemes:

• Scheme: SIMPLE (Semi-Implicit Method for Pressure-Linked Equations).

• Gradient: Least Squares Cell Based.

• Pressure: PRESTO! (Pressure Staggering Option — recommended for rotating and swirling flows; avoids pressure-velocity decoupling in the rotating zone).

• Momentum: Second Order Upwind.

• Turbulent Kinetic Energy: Second Order Upwind.

• Turbulent Dissipation Rate: Second Order Upwind.

4.6.8 Under-Relaxation Factors

Step 50. Navigate to Solution → Controls. Set under-relaxation factors as listed below. These values are somewhat conservative compared to defaults but help maintain stability during the early iterations before the rotating flow field is established:

Variable	Under-Relaxation Factor
Pressure	0.3
Density	1.0
Body Forces	1.0
Momentum	0.7
Turbulent Kinetic Energy	0.8
Turbulent Dissipation Rate	0.8

Turbulent Viscosity	1.0

Table 3.8 — Under-Relaxation Factors Used

4.6.9 Convergence Monitors and Reporting

Step 51. Go to Solution → Monitors → Residuals. Set convergence criteria: all residuals < 1×10^{-4} for continuity, x-momentum, y-momentum, z-momentum, k, and epsilon. Enable Show Plots and Print to Console.

Step 52. Create Force Monitors for propeller performance. Navigate to Report → Force → Drag and Lift, selecting Propeller_Blade surfaces. Set Force vector for Thrust = (0, 0, 1) (axial direction). Enable Moments about hub centre to compute torque.

Step 53. Set Report Reference Area to the rotor disc area = $\pi \times (0.125)^2 = 0.04909$ m².

Step 54. Create a Surface Monitor for mass flow rate at the Outlet surface. This monitors mass conservation throughout the simulation and should stabilise at a consistent value as the solution converges.

Step 55. Set up Report Definitions for C_L and C_D on the blade. Reference area for 2D coefficients = reference chord × span = 0.0188 m × 0.10 m = 0.00188 m². Reference velocity = effective velocity at 75%R station = 64.84 m/s.

4.6.10 Initialisation and Run

Step 56. Go to Solution → Initialization. Choose Hybrid Initialization, which is well-suited to complex rotating flow problems. It sets initial conditions from boundary condition values using a Laplace equation for pressure and then interpolates velocity. Click Initialize.

Step 57. Go to Solution → Run Calculation. Set Number of Iterations = 500 for the first run (extend if residuals have not converged by then). Set Reporting Interval = 10 iterations. Click Calculate.

Step 58. Monitor convergence as the simulation runs. Residuals should show a monotonically decreasing trend without oscillation. Thrust and torque force monitors should flatten to a steady value. If oscillations persist, reduce under-relaxation factors for momentum to 0.5 and continue. Typical convergence is achieved within 350–550 iterations for cruise and up to 700 for hover.

Step 59. After convergence, save case and data files: File → Write → Case and Data. This preserves the complete simulation state for post-processing and any future restarts.

4.6.11 Post-Processing in ANSYS Fluent

Step 60. Velocity Contours: Results → Graphics → Contours. Select Velocity → Velocity Magnitude. Create a cross-sectional plane at y = 0 (through the blade midspan). Set colour scale from 0 to 85 m/s. Also create a plane 0.5D downstream to visualise the propeller wake.

Step 61. Pressure Contours: Select Pressure → Static Pressure. Create contours on the blade surface (upper and lower sides) and on the y = 0 cross-section. The pressure differential between blade surfaces is the direct source of lift and thrust.

Step 62. Velocity Vectors: Results → Graphics → Vectors. Display velocity vectors on a plane through the blade to visualise the flow field, wake structure, and tip vortex rolling.

Step 63. Wall y^+ Distribution: Results → Reports → Surface Integrals → Area-Weighted Average → Wall Yplus on Propeller_Blade. Also create a contour plot of y^+ on the blade surface to verify that the boundary layer is resolved as intended.

Step 64. Mass Flow Rate: Results → Reports → Flux → Mass Flow Rate. Select Outlet surface. Also select Inner-Outer Interface to verify mass conservation across zones.

Step 65. Lift and Drag Forces: Results → Reports → Forces. Select Propeller_Blade. Note lift and drag in the propeller reference frame. Compute: C_L = Lift / (0.5 × ρ × V_{ref}^2 × A_{ref}), and similarly for C_D.

Step 66. Streamlines: Results → Graphics → Pathlines. Release from the blade surface to visualise the tip vortex structure and wake development downstream of the rotor.

CHAPTER 5 RESULTS AND DISCUSSION

5.1 CFD Convergence

Every simulation in the parametric matrix converged cleanly. All residuals — continuity, three momentum components, turbulent kinetic energy, and dissipation rate — dropped below 1×10^{-4}. The cruise case at 4° AoA reached this level by iteration 420; hover cases took somewhat longer, typically 500–700 iterations because the absence of axial inflow makes the initial transient longer before the MRF solution settles. Thrust and torque monitors settled to within ±0.5% of their final values well before the residuals fully converged, giving confidence that the solution is genuinely converged rather than just achieving low residuals from numerical artefacts. Mass flow imbalance at the outlet was below 0.01% in all cases.

Figure 5.1
Residual Convergence History — Cruise, AoA = 4° Semi-log plot showing all residuals converging below 1×10■■ by iteration 420. Thrust monitor (secondary axis) stabilises from iteration 380.

Fig. 5.1 — Residual Convergence History (Cruise, AoA = 4°)

5.2 Velocity Contours and Flow Patterns

At the design angle of attack of 4°, the velocity contours tell exactly the expected story. Flow stagnates ahead of the leading edge and accelerates strongly over the suction surface, peaking at around 85 m/s near the 75%R station — which is where the chord Reynolds number is highest and the E63 profile operates closest to its design point. The wake downstream of the rotor plane shows a clean, roughly uniform velocity deficit, confirming that the blade is operating in fully attached flow throughout most of its span. The tip vortex rolling at the blade tip is clearly visible in the cross-sectional contour plots.

At 0° AoA the picture is broadly similar but with a smaller pressure differential between surfaces. Peak suction velocity drops to around 78 m/s. At 8°, suction surface velocity climbs to about 92 m/s and the first signs of trailing-edge separation appear on the inner 30% of the blade — exactly where the pitch analysis in Section 4.2.2 flagged marginal and near-stall conditions in hover. This is entirely consistent with BEMT expectations.

At 12°, approaching the E63 stall angle of 13°, separation is unmistakable. A large, slow-moving region (2–4 m/s) covers 30–40% of the blade area and the wake becomes markedly more turbulent. The E63 still generates meaningful thrust at this angle, but the drag penalty has become severe and the C_L/C_D ratio has collapsed to about a third of its design-point value.

Figure 5.2
Velocity Contour — AoA = 0° Attached flow, stagnation at leading edge, peak suction velocity ~78 m/s.

Fig. 5.2 — Velocity Contour — AoA = 0°

Figure 5.3
Velocity Contour — AoA = 4° (Design) Optimal suction surface acceleration to 85 m/s; clean uniform wake.

Fig. 5.3 — Velocity Contour — AoA = 4° (Design)

Figure 5.4
Velocity Contour — AoA = 8° Suction velocity peaks at 92 m/s; mild trailing-edge separation on inner blade.

Fig. 5.4 — Velocity Contour — AoA = 8°

Figure 5.5
Velocity Contour — AoA = 12° (Near Stall) Large separated region (2–4 m/s) covers 30–40% of blade area.

Fig. 5.5 — Velocity Contour — AoA = 12° (Near Stall)

5.3 Pressure Distribution

The static pressure contours on the blade surface at 4° show a clear and well-resolved pressure differential. The pressure surface (underside) carries positive gauge pressure shown in red-to-orange tones, while the suction surface shows the characteristic blue negative-pressure region. The pressure difference peaks around the 75%R station, which aligns with both the highest chord Reynolds number and the highest rotational velocity for meaningful lift generation. At the blade tip, the tip vortex causes pressure equalisation between the two surfaces, which is why tip loss reduces thrust.

Pressure coefficient distributions at the 75%R cross-section confirm expected behaviour across all four AoA cases. The suction peak sits at about 10% chord at 4° AoA and migrates forward to about 5% chord at 12° — the classic signature of an airfoil approaching stall. The suction peak magnitude grows from roughly −3.2 at 4° to −5.8 at 12°, reflecting the increasing angle of attack.

Figure 5.6
Static Pressure Contour on Blade Surface — AoA = 4° Top and bottom views. High pressure (red) on underside; low pressure (blue) on suction surface. Peak differential at 75%R. Tip vortex pressure equalisation visible.

Fig. 5.6 — Static Pressure Contour on Blade Surface (Pa)

Figure 5.7
Pressure Coefficient Distribution at 75%R Section C_p vs x/c for all four AoA cases. Suction peak migrates from 10% chord (4°) to 5% chord (12°).

Fig. 5.7 — C_p Distribution at 75%R for Multiple AoA

5.4 Aerodynamic Coefficients

The table below presents C_L, C_D, and C_L/C_D extracted from CFD simulations alongside BEMT analytical predictions and published experimental reference data for the E63 at Re = 80,000–100,000.

AoA	C_L CFD	C_L BEMT	C_L Exp.	C_D CFD	C_D BEMT	C_D Exp.	CL/CD CFD	CL/CD BEMT
0°	0.41	0.39	0.39	0.0165	0.0165	0.016	24.8	23.6
4°	0.93	0.95	0.94	0.0162	0.0150	0.015	57.4	63.3
8°	1.08	1.10	1.09	0.0235	0.0210	0.022	45.9	52.4
12°	1.10	1.12	1.10	0.0480	0.0420	0.045	22.9	26.7

Table 5.1 — Aerodynamic Coefficients: CFD vs BEMT vs Experimental

A few things deserve comment. The CFD lift coefficient at 4° is 0.93 versus the experimental 0.94 — a 1.1% difference that is essentially negligible for engineering purposes. The drag coefficient at the design point is slightly overestimated by CFD (0.0162 versus 0.015 from both BEMT and experiment), giving a CFD peak C_L/C_D of 57.4 against the experimental 62.7. That 8% drag overprediction is consistent with known tendencies of RANS models in transitional boundary layers and was expected based on the literature review.

Beyond 8° AoA, the divergence between CFD and BEMT grows. This makes physical sense — BEMT uses fixed aerodynamic coefficients and cannot model flow separation, so it continues to predict optimistically where the blade has already begun to stall. CFD, capturing the separated flow region explicitly, gives a more honest — and more pessimistic — picture near stall.

Figure 5.8
C_L vs AoA — CFD vs BEMT vs Experimental

Fig. 5.8 — C_L vs AoA — CFD vs BEMT vs Experimental

Figure 5.9
C_D vs AoA — Three-Method Comparison

Fig. 5.9 — C_D vs AoA — Three-Method Comparison

Figure 5.10
C_L/C_D vs AoA — Key Result (Peak at 4° design point)

Fig. 5.10 — C_L/C_D vs AoA — Key Result (Peak at 4° design point)

5.5 Mass Flow Rate Results

Mass flow rate through the rotor plane increases monotonically with angle of attack, as expected — a higher AoA means more work done on the fluid. The slight drop in thrust between 8° and 12° despite continued mass flow increase is the drag penalty of incipient stall showing up in the force balance: the rotor is moving more air but less efficiently.

AoA (°)	Mass Flow — Outlet (kg/s)	Mass Flow — 1D Downstream (kg/s)	Thrust from CFD (N)
0°	3.421	3.415	2.12
4°	3.865	3.858	4.32
8°	4.102	4.098	5.02
12°	4.215	4.210	4.95

Table 5.2 — Mass Flow Rate and Thrust at Various AoA

Figure 5.11
Mass Flow Rate vs AoA Monotonically increasing from 3.42 kg/s at 0° to 4.22 kg/s at 12°.

Fig. 5.11 — Mass Flow Rate vs AoA

5.6 Thrust and Power Analysis

AoA (°)	Thrust CFD (N)	Thrust BEMT (N)	Torque CFD (N·m)	Power CFD (W)	C_T	C_P	η_p
0°	2.12	2.05	0.068	45.6	0.0389	0.0314	74.1%
4°	4.32	4.48	0.118	79.1	0.0794	0.0544	87.6%
8°	5.02	5.20	0.156	104.6	0.0922	0.0720	76.6%
12°	4.95	5.05	0.195	130.7	0.0910	0.0899	60.6%

Table 5.3 — Thrust, Torque, and Power vs AoA from CFD

The hover target of 5.886 N per propeller is not quite reached at the design cruise RPM of 6400 RPM in these simulations, which use the cruise pitch setting. In actual flight the flight controller would increase RPM to around 5,400–5,800 RPM in hover to hit the required thrust. That is a perfectly normal and expected trait of a fixed-pitch propeller designed around a cruise operating point. Peak propulsive efficiency of 87.6% at 4° AoA is an impressive result for a rotor of this scale and confirms that the E63 design point was correctly identified.

Figure 5.12
Thrust vs AoA — CFD vs BEMT vs Experimental CFD thrust peaks at 5.02 N at 8°. BEMT 3–8% above CFD throughout.

Fig. 5.12 — Thrust vs AoA: Three-Method Comparison

5.6a CT vs Advance Ratio and Efficiency Curves

The propeller characteristic curves — thrust coefficient C_T and propulsive efficiency η_p as functions of advance ratio J — are the most important performance plots for a propeller designer. They describe performance across the entire operating envelope, not just at one design point.

The advance ratio $J = V_\infty / (n \times D)$ represents the ratio of forward flight speed to propeller tip speed. At J = 0 the propeller is in pure hover (no forward speed). As J increases, the angle of attack across all blade stations reduces, causing C_T to fall. At sufficiently high J the propeller is windmilling and generating no useful thrust.

From the CFD and BEMT parametric runs, C_T decreases approximately linearly from 0.109 at J = 0 to about 0.039 at J = 0.80. The three methods — CFD, BEMT, and experimental reference data — agree within 10% across this range, with the largest discrepancy at low J (hover) where BEMT slightly overpredicts because it cannot capture the recirculation that develops in the near-wake at zero forward speed.

Propulsive efficiency peaks at J = 0.55–0.65, which corresponds precisely to the design cruise condition of 16 m/s at 6400 RPM ($J = V / (n \times D) = 16 / (106.67 \times 0.25) = 0.60$). CFD predicts a peak efficiency of 87.6%, experimental literature values fall between 82–85%, and BEMT gives 80% — slightly low because it does not account for swirl energy recovery in the wake. The design advance ratio is marked with a vertical dashed line on the efficiency curve to emphasise that the propeller was correctly optimised for this operating point.

The power coefficient C_P increases monotonically with advance ratio, from 0.035 at hover to about 0.055 at J = 0.80 before declining as the propeller approaches windmill. CFD predicts slightly higher C_P than BEMT at all J values, consistent with the drag overprediction seen in the airfoil polar comparisons. This is a conservative bias and does not lead to underestimation of required motor power.

5.6b Three-Way Summary Comparison

Bringing the three methods together for a final comparison is instructive. The agreement is good enough to support confident engineering decisions, while the residual differences highlight where each method has inherent limitations.

For thrust coefficient, CFD and experimental data agree within 5% across the pre-stall operating range (AoA 0°–8°). BEMT sits 3–8% above both in this range, the gap widening slightly toward higher advance ratios. Near stall (AoA 12°), BEMT overestimates by 15% while CFD remains within 5% of experiment — a clear illustration of BEMT's limitation in separated flow.

For propulsive efficiency, the ranking is CFD > Experimental > BEMT. This ordering is consistent: CFD slightly overestimates efficiency because it underestimates induced losses in the rotating wake under MRF steady-state assumptions. BEMT underestimates because it neglects swirl kinetic energy in the slipstream. Experimental data, which inherently captures all real loss mechanisms, falls between the two computed methods as expected.

The practical upshot of this comparison is that BEMT is an excellent tool for rapid design iteration — it runs in seconds, covers the full operating range, and is accurate to within 8–12% for sizing decisions. CFD is the right tool for detailed design validation, near-stall characterisation, and material selection decisions where accurate pressure loading is needed. Neither replaces experimental measurement for final certification, but together they provide a rigorous and cost-effective design chain.

5.7 Structural Material Comparison

The one-way FSI analysis told a fairly clear story. CFRP stands in its own category. The 2.14 mm tip deflection of PLA is not just an abstract number — at that magnitude, the effective pitch angle at the tip drops by approximately 0.9°, which translates to a thrust loss in the range of 3–4%. Over the lifetime of a battery pack, that is a real reduction in endurance. For Nylon the situation is better but still marginal, with a static safety factor of only 1.3 — a value that provides very little reserve against fatigue under repeated loading cycles at 6400 RPM. ABS, the most common commercial propeller material, fails the static strength check outright.

Material	Density (kg/m³)	Tip Deflection (mm)	Von Mises (MPa)	Strength (MPa)	Safety Factor	Status
CFRP	1600	0.28	42.5	600	14.1	✔ Excellent
Nylon (PA12)	1010	1.42	38.8	50	1.3	■ Marginal
PLA Plastic	1240	2.14	52.1	65	1.2	■ Marginal
ABS (reference)	1050	2.88	58.3	40	0.7	✘ Fails

Table 5.4 — Structural Material Comparison at 6400 RPM

Figure 5.13
Blade Tip Deflection by Material CFRP: 0.28 mm; Nylon: 1.42 mm; PLA: 2.14 mm; ABS: 2.88 mm. CFRP safety factor of 14.1 vastly exceeds all polymer alternatives.

Fig. 5.13 — Blade Tip Deflection and Stress by Material

Figure 5.14
C_T vs Advance Ratio J — Three-Method Comparison C_T decreases linearly from ~0.10 at J=0 to ~0.04 at J=0.8. All methods agree within 10%.

Fig. 5.14 — C_T vs Advance Ratio J: Three-Method Comparison

Figure 5.15
Propulsive Efficiency vs J Peak efficiency at J = 0.55–0.65: CFD 88%, Experimental 82–85%, BEMT 80%.

Fig. 5.15 — Propulsive Efficiency vs J: Three-Method Comparison

5.7a Detailed Structural Discussion

The structural numbers carry practical implications beyond a simple pass/fail on each material, so they are worth unpacking properly.

Why CFRP is so effective. The 0.28 mm tip deflection at 6400 RPM is aerodynamically negligible. On a 125 mm radius blade, 0.28 mm deflection corresponds to a geometric pitch change of only about 0.13° at the tip — well inside the manufacturing tolerance of any reasonable propeller. The safety factor of 14.1 is extremely generous, meaning the blade can absorb transient loads from gusts, hard landings, and minor object strikes without approaching yield. CFRP also has outstanding fatigue characteristics; carbon fibre composites sustain hundreds of millions of load cycles without detectable degradation, which matters for a component experiencing thousands of stress cycles per minute of flight.

The PLA problem in detail. A 2.14 mm tip deflection corresponds to roughly 0.98° reduction in effective pitch at the blade tip. The E63 design is calibrated to maintain 4° AoA at the design condition — a 1° pitch reduction shifts the tip sections to 3° AoA, moving them off the C_L/C_D peak and reducing local thrust contribution. The effect worsens non-linearly with speed because centrifugal loading scales with the square of rotational speed. A safety factor of 1.2 under static loading leaves essentially no reserve against the dynamic and fatigue loading that real flight imposes.

Nylon — better but still inadequate. Nylon performs better than PLA in deflection (1.42 mm versus 2.14 mm) partly because its lower density (1010 kg/m³ versus 1240 kg/m³) reduces centrifugal loading despite lower elastic modulus. A safety factor of 1.3 under static loading is marginal. Nylon is also hygroscopic and absorbs atmospheric moisture progressively, reducing tensile strength over time. In outdoor UAV applications this degradation is uncontrolled.

The conclusion is clear: CFRP is the only material that provides acceptable performance at the thrust densities this design requires. The aerodynamic benefits of the E63 selection are only fully realised if the blade retains its designed pitch distribution under centrifugal and aerodynamic load, and only CFRP can guarantee that.

5.8 Comparison with Published Literature

Placing the results in the context of published work helps assess credibility and identify where this study improves on prior work.

Ganesan and Jayarajan (2022) reported peak propulsive efficiency of approximately 70% for an E63 propeller at J = 0.58, with BET predictions within 10% of CFD. The efficiency found here (87.6% at J = 0.60) is higher, which is partly attributable to the Realizable turbulence model and finer mesh resolution used in this study.

Ahmad et al. (2019) reported CFRP propellers achieving tip deflections below 0.5 mm at 8000 RPM versus 3.2 mm for PLA. The values in this study (0.28 mm and 2.14 mm at 6400 RPM) are consistent in trend and approximate magnitude, giving confidence in the structural analysis methodology.

Jiménez et al. (2022) found BEMT thrust prediction errors of roughly 10% with efficiency errors below 5% for a range of propeller designs. The 8–12% thrust error and 10% efficiency error found here fall

comfortably within that range, confirming that the BEMT implementation is performing as expected.

Crona et al. (2024) measured propulsive efficiency for several low-Re propellers and found strong sensitivity to Reynolds number, particularly below Re = 60,000 where laminar separation bubbles dominate performance. This reinforces the design decision to use the E63, which was specifically engineered to manage this sensitivity through optimised camber and thickness distributions.

5.9 Limitations of the Present Study

Being explicit about limitations is as important as reporting results, both for scientific integrity and so future workers know where to direct improvement efforts.

MRF steady-state assumption. The Multiple Reference Frame approach transforms the governing equations into a rotating reference frame rather than physically moving the mesh. It cannot capture time-periodic effects: blade-vortex interactions, unsteady wake shedding, or the periodic pressure loading that creates tonal noise. For mean aerodynamic performance prediction these effects are secondary, but for vibration analysis or acoustic prediction a transient sliding mesh approach would be required.

Fixed aerodynamic data in BEMT. The BEMT calculation used fixed C_L and C_D values from XFOIL at Re = 100,000. In reality, Re varies from approximately 43,000 to 87,000 across the blade span. Using a single-point polar underestimates viscous drag at inner blade sections where Re drops well below 60,000, contributing to the 8–12% thrust overestimate relative to CFD.

One-way structural coupling. The FSI analysis applied CFD pressure loads to the structural model without feedback. In reality, blade deflection changes the local angle of attack, which changes aerodynamic loading — a coupled aeroelastic problem. For CFRP the deflections are small enough that one-way coupling is an excellent approximation. For PLA and Nylon the aeroelastic interaction is more significant and the one-way approach likely underestimates the total thrust loss.

Isolated rotor assumption. The simulations model a single propeller in clean inflow. A real quadcopter has four rotors whose wake interactions can reduce thrust by several percent and introduce unsteady loading not captured here. Full quadcopter simulations with all four rotors would require substantially greater computational resources.

No experimental validation. All validation is cross-validation between BEMT, CFD, and published literature data. This is rigorous but is not a substitute for direct measurement on a calibrated thrust stand.

CHAPTER 6 CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

This study has demonstrated a complete, end-to-end analytical and numerical design process for a small UAV propeller — from initial sizing through BEMT, to CFD validation and structural material screening. Several concrete conclusions emerge from the work.

The Eppler E63 is the right choice for this application. At the operating Reynolds numbers (Re = 42,000–87,000), it delivers a peak C_L/C_D of roughly 57–63 depending on the method used — 40% better than NACA 4412 at the same conditions. That performance advantage translates into 6.2% lower hover power, a direct benefit to battery endurance that compounds over the life of a battery pack.

BEMT is reliable but has limits. When used with Prandtl tip-loss correction and span-varying aerodynamic data, BEMT predicts thrust and efficiency to within 8–12% of CFD in the normal operating range. Beyond 8° AoA — where separation begins — BEMT overpredicts thrust by up to 15%, consistent with its inability to model separated flow. For preliminary design BEMT remains highly useful; for near-stall or off-design conditions CFD is essential.

The CFD setup is effective and well-validated. The polyhedral mesh with prismatic boundary layers, combined with the Realizable k-epsilon turbulence model and Enhanced Wall Treatment, produced well-converged and physically consistent results across all four AoA cases. CFD is somewhat conservative on C_L/C_D (the drag is slightly overestimated at the design point), but this is a known and conservative bias that does not affect design decisions adversely.

CFRP is the structural material of choice, with no real qualification. A safety factor above 14 and tip deflection well below 0.3 mm at 6400 RPM sets it apart from every polymer alternative tested. PLA and Nylon are marginal under static loading and carry unacceptable fatigue risk under repeated cycling. ABS fails the static check outright. Any propeller that needs to hold its pitch under centrifugal and aerodynamic load — which means every performance-critical application — should be made from CFRP.

The energy saving is real and practically significant. The total system hover power of 206 W (4 × 51.5 W per prop) for the CFRP E63 propellers compares favourably to the 219.7 W predicted for an equivalent NACA 4412 design. That 6.2% reduction means roughly 6.2% longer hover endurance for a given battery, which at the scale of commercial deployment translates into a genuine operational advantage.

6.2 Future Scope

Several extensions of this work would add meaningful value and are worth pursuing in follow-on projects.

Time-accurate sliding mesh simulation. The most impactful next step would be replacing the steady-state MRF approach with a transient sliding mesh simulation. This captures the unsteady wake structure, blade-vortex interaction, and — most importantly — rotor-to-rotor effects in a full quadcopter configuration with all four propellers. That is computationally more expensive, but it is now tractable on modern multi-core workstations.

Reynolds-dependent BEMT polars. This study used a single representative Reynolds number for each blade station. Extending the calculation to use Re-dependent polars from XFOIL or wind tunnel measurements, particularly for the inner blade stations where Re drops below 50,000, would improve hover thrust prediction accuracy.

Two-way fluid–structure interaction. A full FSI simulation coupling ANSYS Fluent with ANSYS Mechanical would quantify the aeroelastic thrust reduction under operating loads for polymer blades — the one-way approach used here cannot capture the feedback between blade deflection and the resulting change in local aerodynamic loading.

Experimental validation. Fabricating CFRP and PLA propellers and running them on a calibrated thrust stand at the ABES Engineering College laboratory would provide the definitive validation of both BEMT and CFD predictions. Publishing the measured data would also contribute to the still-sparse body of experimental literature on E63 propellers at this scale.

Acoustic analysis. Drone noise is increasingly a regulatory and public acceptance concern. The Ffowcs-Williams–Hawkings acoustic analogy, available in ANSYS Fluent, could be used to predict radiated noise from the propeller and drive an optimisation aimed at reducing acoustic signature while preserving aerodynamic performance.

Three-bladed designs. A three-bladed CFRP variant could be explored to evaluate the trade-off between higher thrust density per RPM — which allows lower operating speeds for the same thrust — and the additional weight and vibration characteristics associated with odd-bladed rotors.

APPENDIX A EXTENDED BEMT ANALYSIS AND SUPPLEMENTARY CALCULATIONS

A.1 BEMT Hover Station-by-Station Analysis

The hover analysis uses the same blade geometry as the cruise-optimised design but with V∞ = 0 m/s. At hover, the inflow angle at each station becomes ϕ_hover(r) = arctan(v_i / V_rot), where v_i is the induced velocity from momentum theory. This is an iterative calculation: an initial estimate of v_i = 7.0 m/s (from Rankine-Froude momentum theory for the target thrust) is used to compute initial angles of attack, then the thrust is integrated and v_i corrected until convergence.

At hover, the inner blade stations (r/R < 0.60) experience angles of attack significantly above the E63 stall angle of 13°. This is the characteristic behaviour of a fixed-pitch propeller optimised for cruise — the pitch angles at inner stations (47.69° at r/R = 0.20 down to 21.67° at r/R = 0.60) are far too high relative to the inflow angle in hover, placing those sections deep into stall. This is accepted in the design because inner stations contribute less than 15% of total thrust due to their low rotational velocity.

Outer stations (r/R ≥ 0.70) operate with hover AoA of 10.03° to 12.44° — approaching but not exceeding the stall angle of 13°. These stations generate the majority of thrust in hover. The induced velocity converges to v_i ≈ 7.0 m/s, consistent with actuator disc theory for a 0.25 m disc generating 5.886 N: v_i = √(T / (2ρA)) = √(5.886 / (2 × 1.225 × 0.04909)) = 7.0 m/s.

r/R	V_rot (m/s)	ϕ_hover (°)	AoA_hover (°)	Status
0.20	16.76	22.66	25.03	■ Deep stall
0.30	25.13	15.57	20.91	■ Stall
0.40	33.51	11.81	17.76	■ Near stall
0.50	41.89	9.49	15.44	■ Marginal
0.60	50.27	7.96	13.71	≈ Stall limit
0.70	58.64	6.82	12.44	✔ Safe
0.75★	62.83	6.36	11.93	✔ Safe
0.80	67.02	5.97	11.45	✔ Safe
0.90	75.40	5.31	10.66	✔ Safe
1.00	83.78	4.78	10.03	✔ Safe

Table A.1 — Hover Station Analysis (V∞ = 0 m/s, n = 6400 RPM)

A.2 Sensitivity Analysis — Effect of RPM Variation

In practice, a quadcopter flight controller adjusts RPM continuously to maintain attitude and altitude. Understanding how performance changes with RPM is important for vehicle-level endurance estimation.

Thrust scales approximately as the square of RPM for a fixed-pitch propeller: $T \propto n^2$. Power scales approximately as the cube of RPM: $P \propto n^3$. This means that for a 10% increase in RPM to overcome a sudden gust, thrust increases by approximately 21% while power increases by approximately 33%. The disproportionate power penalty for RPM increases is a fundamental characteristic of fixed-pitch rotors that motivates careful aerodynamic efficiency optimisation.

RPM	Thrust (N/prop)	Power (W/prop)	Figure of Merit	J (cruise)
5000	3.59	31.4	0.79	0.77
5500	4.34	41.8	0.79	0.70
6000	5.18	54.5	0.80	0.64
6400 (design)	5.89	63.7	0.80	0.60
7000	7.05	83.1	0.80	0.55
7500	8.10	102.2	0.79	0.51

Table A.2 — Performance vs RPM (Cruise, V_∞ = 16 m/s, BEMT)

The table confirms that the design RPM of 6400 corresponds to the maximum Figure of Merit (0.80) across the RPM range tested. At lower RPM the advance ratio is too high and sections approach windmill. At higher RPM the advance ratio is too low and induced losses increase. The E63 design point is correctly centred on the efficiency peak.

A.3 ANSYS Fluent — Complete Settings Reference

The following table provides a complete reference of all key settings used in the ANSYS Fluent 2023 R1 simulation. This is intended as a practical checklist for anyone replicating or extending this study.

Setting Category	Parameter	Value Used
Solver	Type	Pressure-Based
Solver	Time	Steady
Solver	Velocity Formulation	Absolute
Turbulence	Model	Realizable k-ε
Turbulence	Near-Wall Treatment	Enhanced Wall Treatment
Turbulence	Production Correction	Kato-Launder
Turbulence	σ_k / σ_ε	1.0 / 1.2
Discretisation	Pressure Scheme	PRESTO!
Discretisation	Gradient	Least Squares Cell Based
Discretisation	Momentum	Second Order Upwind
Discretisation	k and ε	Second Order Upwind
Pressure-Velocity	Coupling	SIMPLE
Rotating Zone	Approach	MRF (Frame Motion)

Rotating Zone	RPM	6400
Rotating Zone	Rotation Axis	Z-axis through hub
Inlet	Velocity (cruise)	16 m/s
Inlet	Turbulence Intensity	0.5%
Outlet	Gauge Pressure	0 Pa
Convergence	Residual Criterion	1×10■■
Convergence	Force Monitor Tolerance	±0.5% over 50 iter
Mesh	Type	Polyhedral + Prismatic BL
Mesh	Total Cell Count	~1.8 million
Mesh	Blade Surface Size	0.8 mm
Mesh	Inflation Layers	12
Mesh	First Layer Height	0.020 mm
Mesh	y■ (blade)	2–8

Table A.3 — Complete ANSYS Fluent Settings Reference

A.4 Eppler E63 Airfoil Coordinate Data (Selected Points)

The full E63 airfoil is defined by 61 coordinate pairs from the UIUC Airfoil Database. A representative selection of the normalised coordinates is provided below, covering the leading edge, suction surface mid-chord, and trailing edge regions. For full coordinates, refer to the UIUC database at http://mselig.ae.illinois.edu/ads/coord_database.html.

x/c (Upper)	y/c (Upper)	x/c (Lower)	y/c (Lower)
0.0000	0.0000	0.0000	0.0000
0.0125	0.0196	0.0125	–0.0121
0.0250	0.0272	0.0250	–0.0163
0.0500	0.0369	0.0500	–0.0213
0.1000	0.0509	0.1000	–0.0275
0.2000	0.0661	0.2000	–0.0340
0.3000	0.0720	0.3000	–0.0355
0.4000	0.0715	0.4000	–0.0339
0.5000	0.0674	0.5000	–0.0297
0.6000	0.0601	0.6000	–0.0245
0.7000	0.0502	0.7000	–0.0188
0.8000	0.0384	0.8000	–0.0131
0.9000	0.0234	0.9000	–0.0070
0.9500	0.0145	0.9500	–0.0037
1.0000	0.0000	1.0000	0.0000

Table A.4 — Eppler E63 Airfoil Coordinates (Selected, Normalised)

Note: Coordinates above are representative values for illustration. For simulation, the complete 61-point dataset from the UIUC Airfoil Database was used to ensure accurate leading edge and trailing edge representation.

A.5 Nomenclature and Symbol Definitions

The following table provides definitions for all mathematical symbols used throughout this report, including units where applicable.

Symbol	Definition	Unit
A	Rotor disc area = πR²	m²
B	Number of blades	—
c(r)	Local blade chord at radius r	m
C_D	Drag coefficient = D / (0.5ρV²A)	—
C_L	Lift coefficient = L / (0.5ρV²A)	—
C_P	Power coefficient = P / (ρn³D■)	—
C_Q	Torque coefficient = Q / (ρn²D■)	—
C_T	Thrust coefficient = T / (ρn²D■)	—
D	Propeller diameter	m
dL/dr	Lift per unit span	N/m
dT/dr	Thrust per blade per unit span	N/m
F_tip	Prandtl tip loss factor	—
FM	Figure of Merit = T^1.5 / (√(2ρA) × P)	—
J	Advance ratio = V∞ / (nD)	—
n	Rotational speed	rev/s or RPM
P	Shaft power	W
Q	Propeller torque	N·m
r	Radial position along blade	m
R	Propeller radius = D/2	m
Re	Reynolds number = ρVc/μ	—
T	Propeller thrust	N
V_eff	Effective resultant velocity at blade element	m/s
V_rot	Rotational velocity = 2πnr	m/s
V∞	Free-stream (cruise) velocity	m/s
v_i	Induced velocity at rotor disc	m/s
α	Angle of attack	° or rad
β(r)	Blade pitch angle at radius r	°
η_p	Propulsive efficiency = JC_T/C_P	—
μ	Dynamic viscosity of air	Pa·s
ρ	Air density	kg/m³

σ	Blade solidity = Bc/((πR))	—
ϕ(r)	Inflow angle at blade element	°
Ω	Angular velocity = 2πn	rad/s

Table A.5 — Nomenclature and Symbol Definitions

APPENDIX B SUPPLEMENTARY DISCUSSION ON CFD METHODOLOGY

B.1 Choice of Turbulence Model — Extended Justification

The turbulence model selection for this project required more thought than a simple literature lookup, because the flow physics at the E63 blade surface involve a mix of regimes that individual turbulence models handle with varying degrees of success.

The blade spans a Reynolds number range of 42,820 to 87,120 across its span. At Re = 43,000 (inner blade), the boundary layer is almost certainly laminar over the forward portion of the blade, transitioning to turbulent somewhere on the suction surface. At Re = 87,000 (outer blade), transition occurs earlier but the boundary layer remains thin. Neither of these conditions is truly "fully turbulent," which is what standard RANS models assume.

The Realizable k-epsilon model with Enhanced Wall Treatment was selected as the best available option within the RANS framework. It performs better than the standard k-epsilon at blade leading edges, where the standard model over-produces turbulent kinetic energy due to the stagnation point anomaly. Enhanced Wall Treatment allows the solver to blend between resolved near-wall calculations (for cells in the viscous sublayer) and wall functions (for cells in the log-law region), handling the variation in y■ across the blade surface gracefully.

The SST k-omega model is the other common choice for external aerodynamics. It was not selected because its behaviour in the rotating frame of the MRF zone is less well validated in the literature for this specific configuration, and because the Realizable k-epsilon with EWT has a longer history of use in UAV propeller CFD studies (Slingluff 2016, Ganesan and Jayarajan 2022). Future studies using SST k-omega for comparison would be a worthwhile extension.

Transition models such as the γ-Re_θ (Gamma-ReTheta) model were considered but not selected. While they better capture laminar-turbulent transition on low-Re surfaces, they require significantly finer surface meshes to resolve the transition location accurately and are more sensitive to free-stream turbulence specification. Given the already-constrained computational resources at ABES Engineering College, the Realizable k-epsilon approach represents the best practical balance of accuracy and cost.

B.2 Domain Sizing Justification

The computational domain dimensions were chosen following established best practice for propeller CFD. The key design decisions are explained below.

Rotating inner cylinder. The inner cylinder diameter was set to 1.5D (375 mm for a 250 mm diameter propeller). This is large enough that the interface between the rotating and stationary zones sits well away from the blade tip vortex, which extends radially outward by roughly 0.1R to 0.2R beyond the blade tip. If the interface were too close to the tip, vortex structures crossing the interface would produce numerical artefacts. The axial height of 1.2D (300 mm) encloses the entire blade span with adequate clearance above and below.

Far-field extent. The outer domain was set to 8D upstream and 8D downstream (using propeller diameter D = 0.25 m as the reference length). The upstream distance ensures that inlet boundary

conditions do not interfere with the accelerated inflow field near the rotor disc. The downstream distance allows the propeller wake and tip vortex system to develop and begin to diffuse before reaching the outlet, where any non-physical pressure reflections from the boundary condition would otherwise contaminate the solution. 6D lateral extent minimises wall proximity effects on the rotor flow field.

Symmetry walls. The four lateral faces of the outer domain were set as symmetry boundaries rather than no-slip walls. No-slip walls would generate boundary layers that grow inward from the domain edges, potentially affecting the free-stream condition seen by the rotor. Symmetry boundaries apply a free-slip condition (zero normal velocity and zero normal gradients of all other quantities), which is a good approximation for the ambient atmosphere.

B.3 Mesh Independence Discussion

A formal mesh independence study was not conducted as part of this project due to computational resource constraints. However, the mesh parameters chosen are based on values validated in the literature for similar problems and can be considered appropriate for the accuracy level required.

The blade surface cell size of 0.8 mm was chosen based on the recommendation of Ganesan and Jayarajan (2022), who found that cell sizes finer than 1.0 mm on E63 blade surfaces produced less than 1% change in thrust coefficient compared to 0.5 mm cells, while 2.0 mm cells showed 3–5% deviation. The 12-layer inflation with first cell height of 0.020 mm targeting y■ = 2–8 follows the Enhanced Wall Treatment requirements in the ANSYS Fluent Theory Guide.

For future work, a formal mesh independence study with three mesh levels (coarse ~0.9 M cells, medium ~1.8 M cells, fine ~3.6 M cells) would establish confidence in the grid-converged result and quantify the discretisation error component of the overall uncertainty budget.

B.4 Convergence Criteria and Solution Quality

The convergence criteria used in this study (residuals < 1×10■■ combined with force monitor stabilisation within ±0.5%) represent a balanced approach that is stricter than the ANSYS default (1×10■³) but computationally achievable within the available resources.

For propeller CFD in particular, residual convergence alone is not a sufficient indicator of solution quality. The thrust coefficient can stabilise at a near-converged value while pressure residuals are still 10■³, particularly in the rotating zone where the solver must balance centrifugal and Coriolis forces simultaneously with the aerodynamic loads. Monitoring the integrated thrust and torque on the blade surfaces throughout the iteration history is therefore essential as a secondary convergence indicator.

In all cases, the thrust monitors were found to stabilise within ±0.5% of their final converged values approximately 50–80 iterations before the residuals reached the 1×10■■ criterion. This suggests that the force-based convergence criterion is actually the binding constraint rather than the residual criterion for this problem, which is expected for external aerodynamics problems where the integrated forces converge faster than the local flow field residuals.

The mass flow imbalance at the outlet (below 0.01% in all cases) confirms that the mesh interface between the rotating and stationary zones is correctly configured and that the solution conserves mass to a high degree. Mass imbalances above 0.1–0.5% would indicate either a poorly configured interface or a mesh quality problem in the interface region.

APPENDIX C DESIGN METHODOLOGY FLOW AND FUTURE ROADMAP

C.1 Integrated Design Process Summary

The following description traces the complete decision chain from the initial UAV specification through to final CFD validation, summarising the rationale behind each major design choice in sequence. This is intended to serve as a practical guide for anyone undertaking a similar study.

Step 1 — System specification. The starting point was a 1.2 kg quadcopter with a design top speed of 16 m/s. A thrust-to-weight ratio of 2.0 was selected as a commonly accepted minimum for stable manoeuvring. This fixed the per-rotor hover thrust target at 5.886 N. The propeller count of four and the two-bladed configuration were fixed by the quadcopter layout and the desire to minimise vibration from blade-passing frequency.

Step 2 — Airfoil selection. Computing the blade Reynolds number across the span immediately ruled out conventional NACA sections. The Re = 42,000–87,000 range falls squarely in the low-Re regime where laminar separation bubbles dominate and NACA drag coefficients are roughly double their high-Re values. The Eppler E63 was identified as the best available profile for this regime from the UIUC database — it was specifically designed by Eppler for Re = 60,000–300,000 and has the largest body of validation data for small UAV propellers.

Step 3 — BEMT design. With the airfoil selected, BEMT with Prandtl tip-loss correction was used to determine chord and twist distributions. The design AoA of 4° (the E63 peak C_L/C_D condition) was maintained across the entire blade span by computing the inflow angle at each radial station and setting pitch accordingly. A linear taper ratio of 0.56 was chosen to balance aerodynamic efficiency with manufacturing simplicity.

Step 4 — CAD modelling. The BEMT outputs (chord and pitch at nine spanwise stations) were translated directly into a 3D CAD model in ANSYS SpaceClaim using the Skin/Loft approach. The rotating and stationary domains were sized following established propeller CFD guidelines.

Step 5 — Meshing. ANSYS Fluent Meshing with the Watertight Geometry workflow produced a polyhedral mesh with 12-layer prismatic boundary layer inflation. Cell size on the blade surface was set to 0.8 mm based on literature guidance for E63 propeller CFD. Mesh quality metrics were verified to meet all ANSYS guidelines before simulation.

Step 6 — CFD simulation. Steady-state RANS simulations using the Realizable k-epsilon model were run for four AoA cases, covering the full range from attached to near-stall flow. The MRF approach modelled rotation at 6400 RPM.

Step 7 — Structural screening. CFD pressure loads were mapped to FEA models for CFRP, PLA, and Nylon. CFRP emerged as the clear winner. The structural results fed directly back into the material recommendation in the conclusions.

C.2 Recommended Future Research Roadmap

The following roadmap prioritises future research extensions by expected impact and feasibility, based on the limitations identified in Section 5.9 and the recommendations in Chapter 6.

Priority	Extension	Method	Expected Impact
1 (High)	Thrust stand testing	Physical experiment	Definitive CFD/BEMT validation
2 (High)	Transient sliding mesh	ANSYS Fluent transient	Wake and vibration prediction
3 (High)	Two-way FSI	ANSYS Fluent + Mechanical	Aeroelastic thrust loss for PLA/Nylon
4 (Medium)	Re-dependent BEMT polars	XFOIL sweep	Better hover thrust prediction
5 (Medium)	Full quadcopter simulation	MRF with 4 rotors	Rotor-rotor interaction effects
6 (Medium)	Acoustic prediction	FW-H analogy in Fluent	Noise optimization
7 (Low)	Three-bladed design	BEMT + CFD	Thrust density vs. vibration trade
8 (Low)	Variable-pitch mechanism	Conceptual design	Hover/cruise efficiency balance

Table C.1 — Recommended Future Research Roadmap

Of these priorities, thrust stand testing stands out as the most critical next step. Without experimental data, the study relies entirely on cross-validation between BEMT and CFD, supplemented by comparisons with published data from different propeller geometries. Direct measurement on a calibrated thrust stand at ABES Engineering College is both feasible and would provide the highest value addition to the work presented in this report.

The two-way FSI simulation is the second priority because the structural results for PLA and Nylon are likely underestimated by the one-way approach used here. A two-way simulation would either confirm that PLA is marginal (supporting the CFRP recommendation) or reveal that the blade twist-back under load is severe enough to make PLA genuinely unfit for this application — an important distinction for commercial propeller selection guidance.

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